Linearization

The time discretization leads to an algebraic system, in which the nonlinearities are preserved. Before the algebraic system can be solved, linearization of (3.56) needs to be applied to gain a linear algebraic system (LAS). The nonlinearity in (3.56) stems from the convective term ($\vec{u} \cdot \nabla \vec{u}$). The stabilization parameters τ depend further on on the velocity, however this nonlinearity is neglected. The stabilization parameter is in the presented scheme constant for a time step because it depends on the velocities of the last time step. Kaltenbacher [91] summarized the mathematical foundation of the linearization. Codina [31] and Wall [146] performed the linearization in context of stabilized FEM. Denoting i the fluid iteration counter the linearization of the convective term is given by

$$\vec{v}^{i+1} \cdot \nabla \vec{v}^{i+1} \approx \vec{v}^{i} \cdot \nabla \vec{v}^{i+1} + \lambda_1 \left[\vec{v}^{i+1} \cdot \nabla \vec{v}^{i} - \vec{v}^{i} \cdot \nabla \vec{v}^{i} \right] \ . \tag{3.97}$$

For $\lambda_1 = 0$ the fixpoint and for $\lambda_1 = 1$ the Newton-Raphson scheme is obtained. The simulations of this thesis are based on the fixpoint iteration, because it possesses a more stable convergence rate. The Newton-Raphson scheme indeed possesses a higher convergence order, but it is quite sensitive to the initial guess. The performed flow simulations with the fixpoint scheme converged within $3 - 7$ iterations. Thereby, no performance improvement is recognized with the Newton scheme. That is why the fixpoint scheme is utilized.

3.1.3 Validation examples

In order to confirm the correct implementation of the novel flow solver, simulations of several different test setups are performed. Both steady and unsteady flow problems are considered. Unsteady flows are thereby treated with the BDF2 scheme (see Sec. 3.1.2). The validation of BE, CN and the two-step can be found in [60]. Firstly, the steady flow through a channel is treated and the results are compared with the analytic solution. Hereby, two different sets of boundary conditions are tested: a velocity driven set and a pressure driven set. Thereafter, two different flows around a circular obstacle are discussed. One at $Re = 40$ resulting in a steady flow and the other at $Re = 100$ for which an unsteady flow develops, known as the Kármán vortex street. Finally, simulation results of an unsteady flow ($Re = 100$) around a square are compared to results of the FVM solver `FASTEST-3D` and a lattice-Boltzmann solver.

Steady channel flow

The domain of the flow-through channel is shown in Fig. 3.4 together with the mesh, consisting of 16×8 second order quadrilateral elements. The fluid flows from left to right. At the top and at the bottom the walls are fixed. In a fully developed plane flow, the y-velocity vanishes and the incompressible Navier-Stokes equations reduce to

$$\frac{dp}{dx} = \frac{\mu}{\rho} \frac{d^2 v_x}{dy^2} \ . \tag{3.98}$$

(3.98) can be solved analytically with appropriate boundary conditions. In the numerical simulation the vanishing y-velocity is incorporated by homogeneous Dirichlet conditions at the whole boundary. The density and the dynamic viscosity of the fluid are set to

$$\rho = 1.0\,\mathrm{kg/m^3} \qquad \text{and} \qquad \mu = 0.002\,\mathrm{Pa\,s}. \tag{3.99}$$

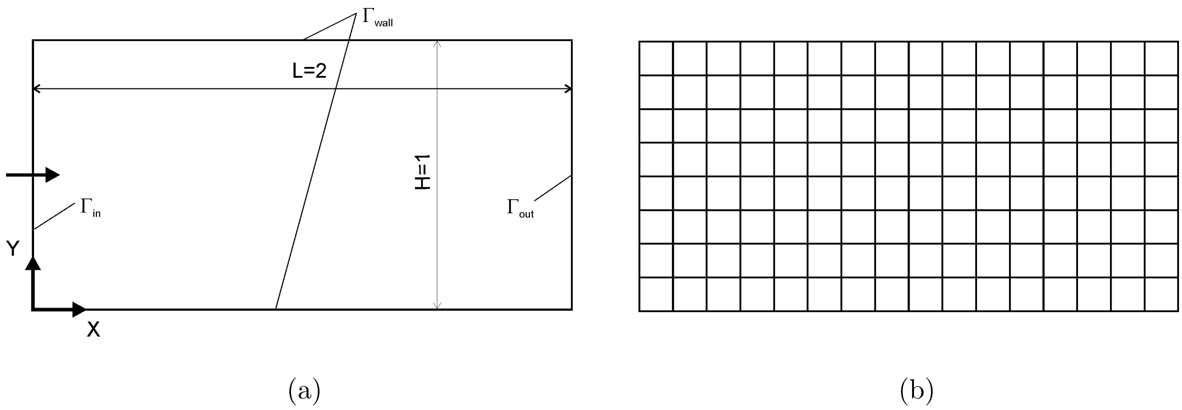


Figure 3.4: Flow-through channel (a) computational domain of the channel (units in m); (b) finite element mesh.

Velocity driven: To obtain a velocity driven flow a parabolic inflow profile is assumed at the inlet boundary Γ_{in}

$$v_x(y) = \frac{1}{2} - \frac{1}{2}\left(\frac{y-0.5}{0.5}\right)^2 \text{ m/s} . \tag{3.100}$$

At the top and at the bottom boundary Γ_{wall}, the x-velocity is set to zero. To ensure the outflow at Γ_{out} a free x-velocity is set. The pressure is set to zero only at one node, the lower right corner node (2 m/0 m). Under these boundary conditions the analytic solution for the pressure is given by

$$p(x) = \frac{P(x)}{\rho} = 8\,\nu\, v_{x\,\max} \frac{L-x}{H^2} \text{ Pa} . \tag{3.101}$$

The numerical results of pressure and x-velocity are shown by contour plots in Fig. 3.5. The analytical and numerical pressure results are consistent, as shown along the horizontal line ($y = 0.5$ m) in Fig. 3.6a.

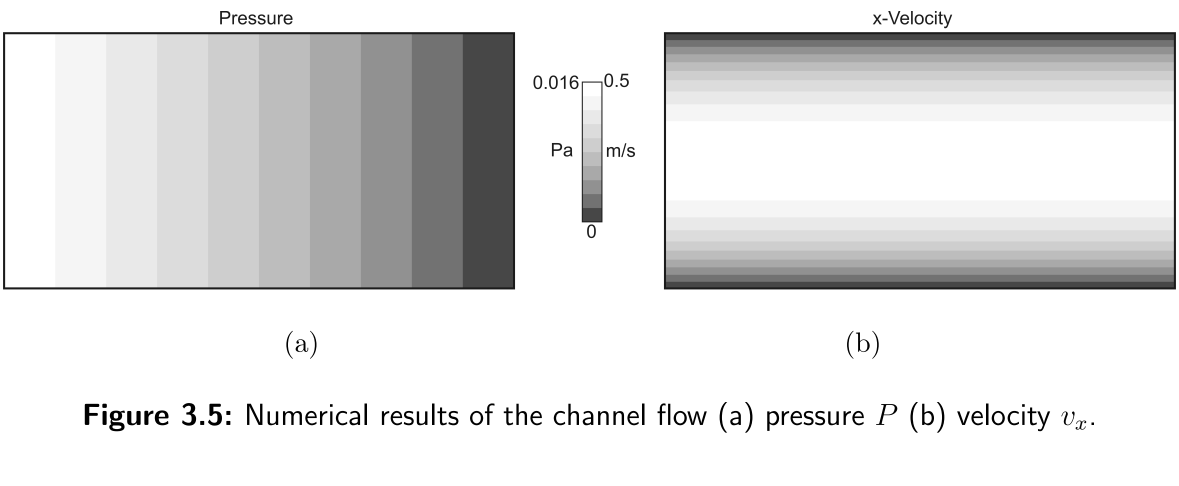
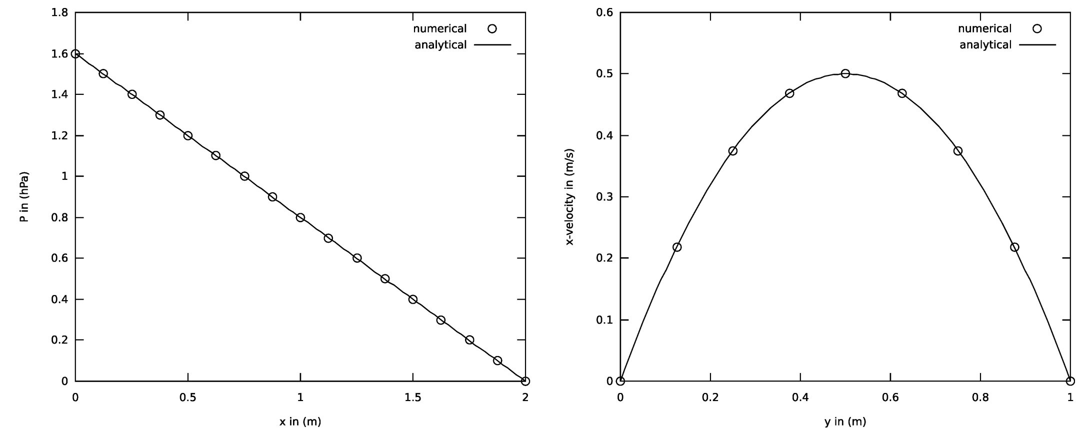


Figure 3.5: Numerical results of the channel flow (a) pressure P (b) velocity v_x.

Pressure driven: The second set of boundary conditions is supposed to prescribe a pressure drop so that the parabolic x-velocity distribution develops. The wall conditions are the



(a) (b)

Figure 3.6: Numerical and analytic results of the channel flow (a) pressure in x-direction along $y = 0.5\,\mathrm{m}$ (b) velocity v_x in y-direction along $x = 1\,\mathrm{m}$.

same as in the velocity driven case. At the in- and outlet the following pressure conditions are applied

$$\vec{h} = P\mathbb{1} \cdot \vec{n} \quad \text{with } P = 0.016\,\mathrm{Pa} \quad \text{on } \Gamma_{\text{in}} \quad \text{and} \quad \vec{h} = 0 \quad \text{on } \Gamma_{\text{out}} \,, \tag{3.102}$$

while the x-velocity is free there. The comparison of the numerically computed x-velocity along the vertical line at $x = 1$ with the analytic solution is shown in Fig 3.6b. The spatial distribution of pressure and velocity is similar to the results depicted in Fig. 3.5. The pressure drop condition is of particular importance for phonation simulations as muscle contraction imposes a pressure drop along the glottis.

A good compliance with the analytical solution is confirmed with both types of boundary conditions.

Flow around a circular obstacle

A more complex and maybe the most famous benchmark in fluid mechanics is given by the flow around a cylinder. The list of literature about measurements and simulations of this setup is quite long; an overview can be found e.g. in [128]. The change from the steady to an unsteady flow around $Re = 50$ is noteworthy. The Kármán vortex street develops in the wake of the cylinder for Reynolds numbers above $Re = 50$. The air flow around an island may have similar flow structures, recognizable as soon as clouds exist, see Fig. 3.7. Two situations are discussed in detail: a steady ($Re = 40$) and an unsteady ($Re = 100$) flow. The geometry of the test case is chosen in accordance with [146] and is shown in Fig. 3.8 together with the finite element mesh. The mesh consist of 7876 Q2 elements.

Steady ($Re = 40$): The steady flow at $Re = 40$ around a cylinder is characterized by two symmetric recirculation eddies in the wake of the cylinder (see Fig. 3.9). The length of these eddies and the location of the flow separation point are famous benchmark results. The density and the dynamic viscosity of the fluid are set to

$$\rho = 1000\,\mathrm{kg/m} \qquad \text{and} \qquad \mu = 4 \times 10^{-4}\,\mathrm{Pa\,s}\,. \tag{3.103}$$